

MIT 18.06SC - 2.3

Problem 16.1

$$C - D = 7$$

$$C + D = 7$$

$$C + 2D = 21$$

$$Ax = b \text{ form: } \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \\ 21 \end{bmatrix}$$

Unsolvable so we solve $A^T A \hat{x} = A^T b$ instead:-

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 7 \\ 7 \\ 21 \end{bmatrix} =$$

$$= \begin{bmatrix} 35 \\ 42 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \text{:- } \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 35 \\ 42 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 & : & 35 \\ 2 & 6 & : & 42 \end{bmatrix}$$

↓

$$\begin{bmatrix} 3 & 2 & : & 35 \\ 0 & 14/3 & : & 56/3 \end{bmatrix}$$

↓

$$\begin{bmatrix} 3 & 2 & : & 35 \\ 0 & 1 & : & 4 \end{bmatrix}$$

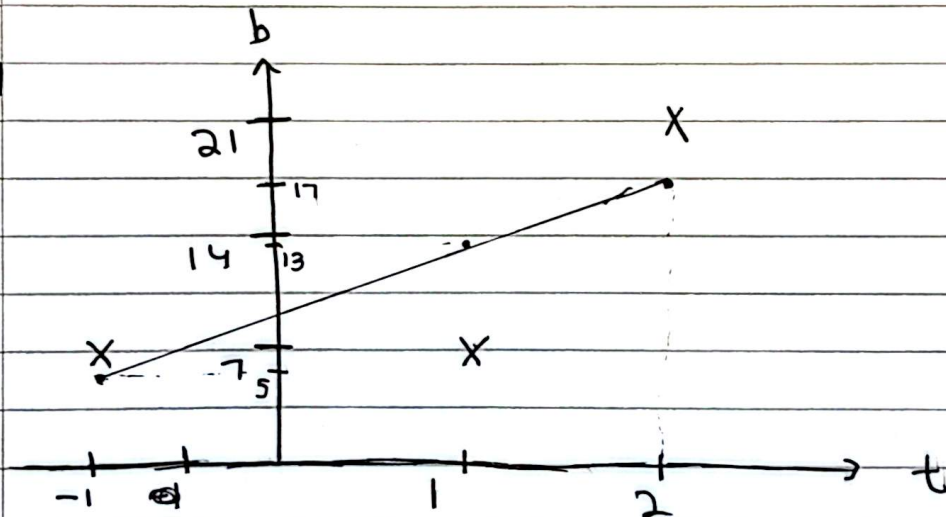
↓

$$\begin{bmatrix} 3 & 0 & : & 27 \\ 0 & 1 & : & 4 \end{bmatrix}$$

↓

$$\begin{bmatrix} 1 & 0 & : & 9 \\ 0 & 1 & : & 4 \end{bmatrix}$$

$$\hat{x} = [9, 4]^T$$



Problem 16.2

$$p = A\hat{x}$$

$$= \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 9 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix}$$

Error vector $\vec{e} = b - p$

$$= \begin{bmatrix} 7 \\ 7 \\ 21 \end{bmatrix} - \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ -6 \\ 4 \end{bmatrix}$$

~~$P_e = 0$ because the error vector is in the nullspace of P .~~

~~$$P_e = P(b - p) = Pb - Pp = 0 \text{ because } Pp = p = 0$$~~

$$\begin{aligned} P_e &= 0 \text{ because } P_e = P(b - p) \\ &= Pb - Pp \\ &= P \cdot 0 \\ &= 0 \end{aligned}$$

Problem 16.2

$$C \neq D = 2$$

$$C + D = -6$$

$$C + 2D = 4$$

$$\text{In } Ax=b \text{ form:- } \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 4 \end{bmatrix}$$

We already know the value of $A^T A$ from ~~the~~ problem 16.1.

$$A^T b = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -6 \\ 4 \end{bmatrix} \\ = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \text{:- } \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 2 \\ 0 & 14/3 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \leftarrow \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

This matrix has full column ~~rank~~ ^{rank} and ~~row~~ ^{rank} therefore the nullspace is only the zero vector so $\hat{x} = [0 \ 0]^T$

$b = (2 \ -6 \ 4)$ is perpendicular to $C(A)$ so the projection $p = 0$.

Problem 16.4

$$C - D = 15$$

$$C + D = 13$$

$$C + 2D = 17$$

$$Ax = b :- \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix}$$

We already know $A^T A$ from problem 16.3. $A^T b$ is:-

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix}$$

$$= \begin{bmatrix} 35 \\ 42 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b :- \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 35 \\ 42 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 3 & 2 & 35 \\ 2 & 6 & 42 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 3 & 2 & 35 \\ 0 & 14/3 & 56/3 \end{array} \right]$$

$$\begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 4 \end{bmatrix} \leftarrow \begin{bmatrix} 3 & 0 & 27 \\ 0 & 1 & 4 \end{bmatrix}$$

$\hat{x} = [9 \ 4]^T$. The error is $e = 0$ because this b is already in the column space of A .

Problem 16.5

The error vector $e \in N(A^T)$. The projection $p \in C(A^T)$. $\hat{x} \in C(A^T)$ and the $N(A) = ce$, where c is any scalar and e is the error vector. is the zero vector only.

Problem 16.6

$$\cancel{C + 4D}$$

$$C - 2D = 4$$

$$C - D = 2$$

$$C = -1$$

$$C + D = 0$$

$$C + 2D = 0$$

$$Ax = b :- \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & -2 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}$$

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ -10 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b :- \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 5 \\ -10 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & : & 5 \\ 0 & 10 & : & -10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & : & 1 \\ 0 & 1 & : & -1 \end{bmatrix}$$

$\hat{x} = \begin{bmatrix} 1 & -1 \end{bmatrix}^T$ i.e. $C=1, D=-1$ is the best fit line